

LED4B-650-0.1

OVERVIEW

Surface LED coupled to a 105/125 multimode optical fiber and packaged into a hermetic case.

MAIN FEATURES

- Wavelength: 650 nm
- Cavity type: Surface LED
- Optical power in CW mode in multimode fiber: typ. 100 μ W
- Package types: coaxial, coaxial with bracket, 14 pins BTF

ORDERING INFORMATION

LED4B-650-0.1-X-X-X-X-X-X

Case type

COAX: compact coaxial (low duty cycle pulse mode only)

COAXB: compact coaxial with a bracket

TH: compact coaxial with a bracket compatible to Thorlabs mount

BTF: 14-pins BTF type 1 (Pump) for active thermal stabilization (TEC and thermistor)

Pinout code

31: see more details on page 4

Fiber type

MM105: MM, [105/125 NA=0.22 \(MM-S105/125-22A\)](#), furcation tubing $\varnothing$ 0.9 mm

Other type on request

Connector type

FU: FC/UPC (MM105)

N: no connector (scissors cut)

Other type: on request

Test measurements

CW: CW mode (electro-optical parameters at $T=25\pm 5$ C and spectrum)

P: Pulse mode (pulse duration 5 us, duty cycle 1%, at $T=25\pm 5$ C)

CWP: both CW and pulse mode

Fiber length

0.5: 500 $\pm$ 50 mm

1.0: 1000 $\pm$ 100 mm

Other length on request

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ABSOLUTE MAXIMUM RATINGS

Parameter		Value	Unit	Conditions
LED forward current	$I_{\max}$	22	mA	CW, $T = 25^{\circ}\text{C}$
LED forward current	I_{pmax}	50	mA	Pulse, 5 μs , duty cycle 1%, $T = 25^{\circ}\text{C}$
LED reverse voltage	V_{RL}	10	V	
Operating temperature*	T_{OP}	-40 - +85	$^{\circ}\text{C}$	Coaxial package
Operating temperature*	T_{OP}	-40 - +60	$^{\circ}\text{C}$	DIL, BTF ($T_{\text{st}} = 25^{\circ}\text{C}$)
Storage temperature	T_{stg}	-40 - +85	$^{\circ}\text{C}$	
Soldering temperature	T_{sold}	260	$^{\circ}\text{C}$	Max. 5 seconds

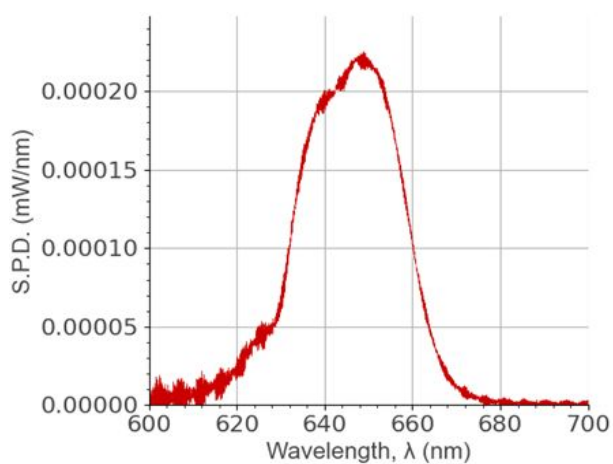
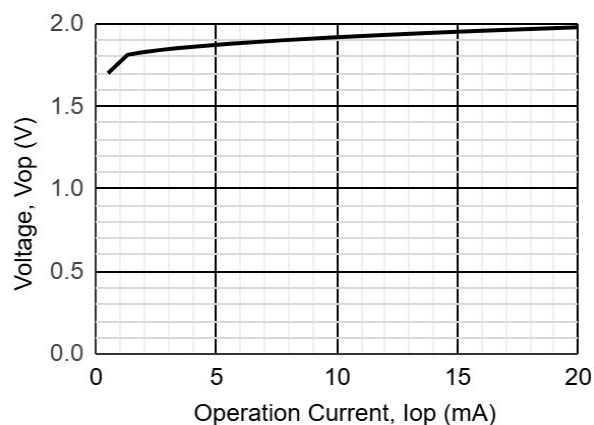
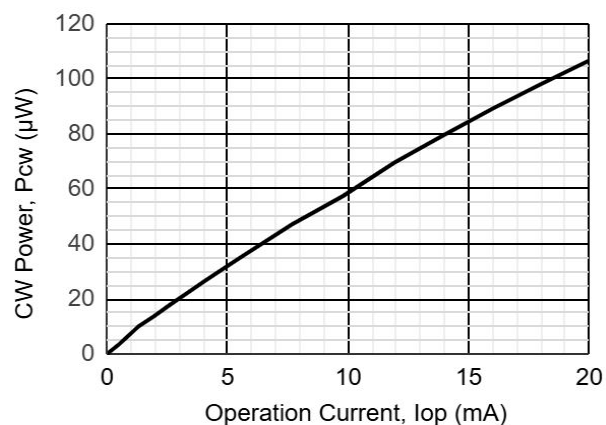
*Operating temperature is defined by the case temperature. It is recommended to ensure sufficient heat dissipation so that the module's maximum operating temperature is not exceeded.

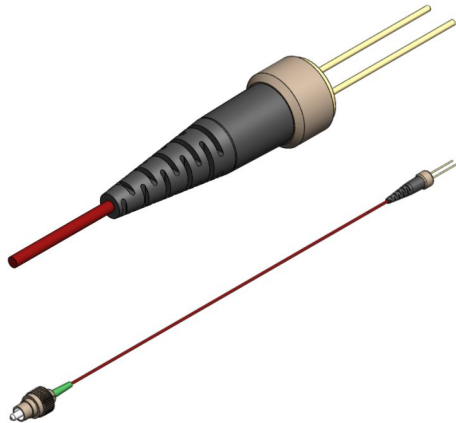
Operating temperature for the BTF 14-pins case T with TEC is defined for internal temperature stabilization at $T_{\text{st}} = 25^{\circ}\text{C}$ that corresponds to thermistor resistance $R_t = 10 \text{ k}\Omega$.

ELECTRICAL-OPTICAL CHARACTERISTICS ($T = 25^{\circ}\text{C}$)

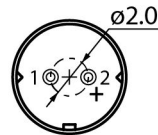
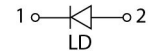
Parameter		MIN	TYP	MAX	Unit	Conditions
Optical power (CW)	P_{CW}	0.05	0.10		mW	CW, $I_{\text{op}} = 20 \text{ mA}$, MM105
Mean wavelength	λ	635	650	660	nm	CW, $I_{\text{op}} = 20 \text{ mA}$
Spectral width	$\Delta\lambda$		25		nm	CW, $I_{\text{op}} = 20 \text{ mA}$
Slope efficiency	S_e		5		$\mu\text{W}/\text{mA}$	CW, MM105
Operating voltage	V_{op}		2.0	2.6	V	CW, $I_{\text{op}} = 20 \text{ mA}$

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LED4B-650-0.1**COAX**

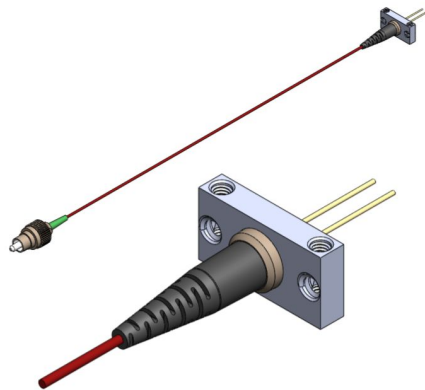
BACK VIEW

PINOUT
#31Download more
information

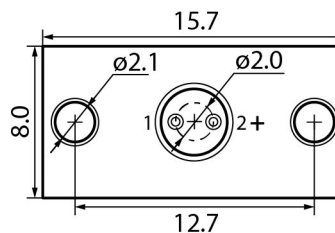
Drawing

3D model

Application Notes

**COAXB**

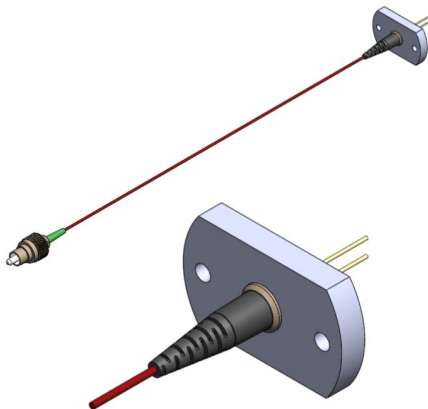
BACK VIEW

PINOUT
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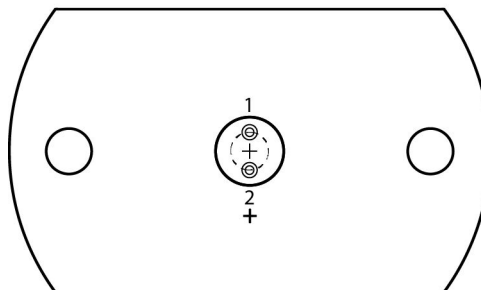
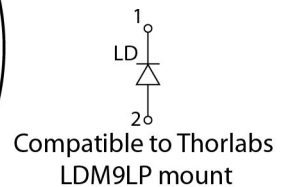
Drawing

3D model

Application Notes

**PACKAGE TH**

BACK VIEW

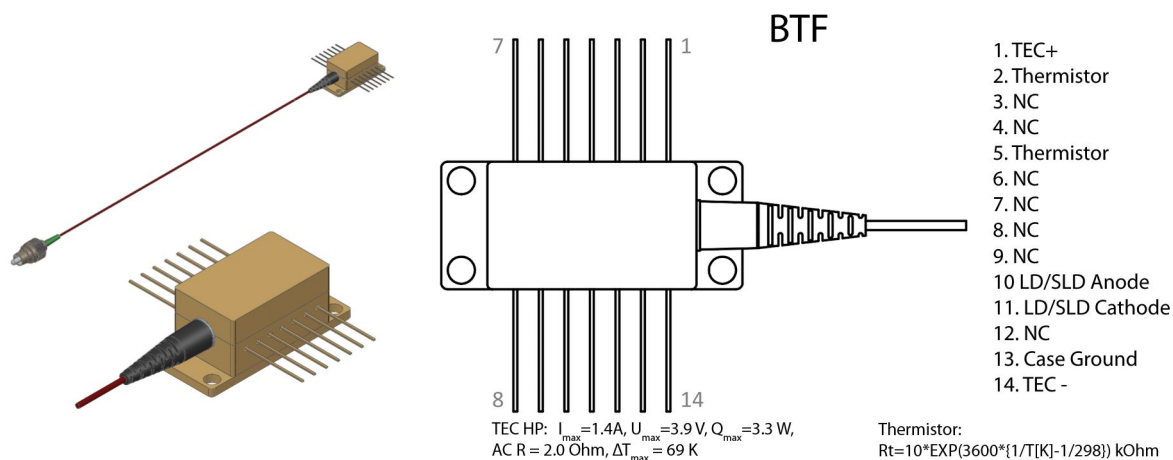
PINOUT
#31
Thorlabs Pin Code GDownload more
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Drawing

3D model

Application Notes

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Drawing



3D model



Application Notes

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Characteristics, data, materials and structures specified in this datasheet are subject to change without notice. Please refer to the latest specification before use of the products.

Safety and handling cautions

1. Avoid smashing and burning of the module. Avoid storing and using the module in conditions where water, organic solvents or aggressive acids or bases may contact the module or where there is a possibility of exposure to corrosive gases, explosive gases, dust, salinity or other harsh conditions. The module should be disposed as special industrial waste.
2. Exceeding absolute maximum ratings even for a short time can cause permanent damage of the module.
3. The module is sensitive to and can be broken by ESD (static electricity).

Conflict Minerals Policy Statement

LD4B achieves business objectives and customer needs with social responsibility. We do not support or contribute to the violence and human rights violations associated with the mining of conflict minerals coming from Conflict Regions according to US "Dodd-Frank Act". When possible, our suppliers' conflict mineral statements are reviewed. We do not directly purchase Conflict Minerals from any source and do not knowingly procure any parts and products containing Conflict Minerals from Conflict Regions.

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